Machine Learning and Optimization for Electricity Market Trading: Uncertainty-Aware Battery Trading Strategies

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***Abstract*—Forecast-driven battery arbitrage requires both accurate electricity price predictions and scheduling decisions that account for operating costs and uncertainty. This paper presents a historical evaluation of machine learning-based day-ahead electricity price forecasting coupled with deterministic, cost-aware, and Conditional Value-at-Risk (CVaR) mixed-integer linear programming for a 1 MW/2 MWh battery. A histogram-based gradient boosting model trained on data through 2015 is evaluated without retraining on German electricity market data for 2019 and 2020.**

Relative to a naive persistence forecast, the model reduces mean absolute error by 21.07% and 22.49%, respectively. ML-based deterministic scheduling increases realized gross trading profit by 12.83% and 10.69% relative to a fixed-schedule benchmark. Under an illustrative high operating-cost scenario, cost-aware optimization increases realized net operating profit by EUR 2,674.50 and EUR 2,747.49. In contrast, stronger CVaR penalization reduces realized gross profit without improving the empirical worst-5% mean daily profit. Furthermore, prediction-error intervals calibrated on historical residuals achieve coverage rates of 84.80% and 82.85%, below their nominal 90% level.

These findings highlight the importance of jointly evaluating forecasting accuracy, economic scheduling performance, and uncertainty calibration under explicit modeling assumptions.

***Index Terms*—Battery energy storage systems, electricity price forecasting, day-ahead electricity markets, machine learning, mixed-integer linear programming, battery arbitrage, stochastic optimization, conditional value-at-risk, battery degradation.**

I. Introduction

The increasing integration of variable renewable energy sources has intensified the need for operational flexibility in electricity systems. Battery energy storage systems (BESS) can provide such flexibility by shifting energy across time and participating in wholesale electricity markets. In particular, day-ahead energy arbitrage enables storage operators to purchase electricity during low-price periods and sell it when prices are higher.

The economic performance of battery arbitrage depends on both the accuracy of electricity price forecasts and the ability to translate these forecasts into feasible and profitable operating schedules. Although electricity price forecasting has been extensively investigated using statistical and machine learning methods, forecasting accuracy alone does not fully characterize the economic value of a predictive model. Forecasting errors can affect scheduling decisions differently depending on their timing, magnitude, and influence on battery operation.

Consequently, forecasting methods should also be evaluated through the economic outcomes of the decisions they support. Previous studies have investigated decision-focused electricity price forecasting for energy storage arbitrage [1], while recent research has examined the relationship between statistical forecasting accuracy and economic performance [2].

Beyond forecasting accuracy, battery operating costs represent an important consideration. Scheduling strategies that maximize gross trading profit may encourage low-margin transactions or excessive cycling when battery degradation and transaction costs are neglected. Previous research has highlighted the importance of incorporating battery aging into electricity market participation [3]. Therefore, the economic evaluation of battery trading strategies should distinguish between gross trading profit and net operating profit under explicitly defined cost assumptions.

Electricity price uncertainty introduces an additional challenge. Stochastic optimization provides a framework for representing alternative future price trajectories, while Conditional Value-at-Risk (CVaR) enables downside-risk considerations to be incorporated into the scheduling objective. However, the effectiveness of risk-aware optimization depends on the quality and representativeness of the underlying scenario distribution. Camuñas García-Miguel et al. [4] showed that incorporating risk measures does not necessarily improve the economic performance of stochastic battery arbitrage compared with deterministic approaches.

These considerations motivate an integrated empirical evaluation of forecast-driven battery trading, in which forecasting accuracy, operating costs, and uncertainty modeling are examined within a consistent optimization framework.

In this study, a machine learning-based electricity price forecasting model is combined with deterministic, cost-aware, and scenario-based stochastic mixed-integer linear programming (MILP) formulations for battery arbitrage in the German day-ahead electricity market. The analysis considers a price-taking 1 MW/2 MWh battery with a round-trip efficiency of 90%.

The forecasting model is trained using historical observations through 2015 and evaluated on subsequent periods, including 2016–2017, 2019, and 2020, without further retraining. Particular attention is given to the realized economic

performance of the resulting schedules, the effects of assumed operating costs, and the transferability of historical forecasting-error distributions.

The main contributions of this study are summarized as follows:

1) *Forecasting accuracy and economic value:* An out-of-sample evaluation of machine learning-based electricity price forecasting and its realized economic value for battery arbitrage, using forecasting and trading benchmarks across multiple historical periods.
2) *Cost-aware battery scheduling:* A comparative assessment of gross-profit and net-profit optimization under consistent battery operating constraints, incorporating illustrative degradation and transaction-cost scenarios.
3) *Risk-aware optimization and uncertainty calibration:* An empirical evaluation of scenario-based stochastic optimization with CVaR, including realized downside-risk metrics and an assessment of the transferability of historical forecasting-error distributions across evaluation periods.

Rather than introducing a new forecasting algorithm or optimization formulation, this study provides a systematic historical comparison of their interactions in battery trading applications. The results indicate that improved forecasting accuracy and explicit operating-cost modeling can enhance realized trading performance under the considered assumptions. Conversely, stronger CVaR penalization does not necessarily improve realized downside-risk outcomes when uncertainty scenarios are constructed from fixed historical error distributions.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the data and forecasting methodology. Section IV presents the battery optimization formulations. Section V describes the experimental setup. Section VI presents and discusses the empirical results, while Section VII concludes the paper and outlines directions for future research.

II. Related Work

A. Electricity Price Forecasting

Electricity price forecasting has been extensively studied using statistical, econometric, and machine learning approaches. Weron [5] provides a comprehensive review of forecasting methodologies and discusses the challenges associated with electricity price prediction.

Lago et al. [6] reviewed state-of-the-art day-ahead electricity price forecasting methods and introduced an open-access benchmark to support reproducible comparisons. Their work emphasizes the importance of appropriate benchmark models, consistent evaluation protocols, and statistical testing when comparing forecasting performance.

These studies provide the methodological basis for chronological out-of-sample evaluation and the comparison of machine learning forecasts against simpler forecasting approaches.

B. Battery Arbitrage and Operating Costs

Battery energy storage systems can exploit temporal electricity price differences through optimized charging and discharging schedules. The economic value of such operation depends on market price variability, storage efficiency, and the operational constraints of the battery.

Mercier et al. [**?**] examined the historical value of electricity storage arbitrage across European day-ahead markets, providing a reference for evaluating arbitrage opportunities under different market conditions.

Beyond gross trading revenue, battery degradation and transaction costs influence the profitability of storage operation. Xu et al. [3] investigated the incorporation of battery cycle-aging costs into electricity market participation, highlighting the importance of accounting for degradation when determining optimal battery dispatch schedules.

These findings motivate the explicit distinction between gross-profit maximization and cost-aware scheduling under defined operating-cost assumptions.

C. Stochastic Optimization and Risk

Electricity price uncertainty introduces additional challenges for battery scheduling, as decisions based on forecasted prices may lead to unfavorable realized outcomes.

Stochastic optimization addresses uncertainty by considering multiple possible future price trajectories. Conditional Value-at-Risk (CVaR) provides a measure of downside risk and can be incorporated into optimization objectives. Rockafellar and Uryasev [7] developed a formulation that enables CVaR optimization through auxiliary variables and linear constraints.

In the context of battery arbitrage, Camuñas García-Miguel et al. [4] investigated the effects of risk measures and battery degradation costs on optimal scheduling. Their findings indicate that risk-aware optimization does not necessarily improve economic performance relative to deterministic approaches.

These results highlight the importance of evaluating risk-aware scheduling through realized economic outcomes, rather than relying exclusively on scenario-based objective values.

D. Positioning of the Present Study

Previous research has examined electricity price forecasting, battery arbitrage, operating costs, and risk-sensitive optimization from different perspectives.

Sang et al. [1] investigated decision-focused electricity price forecasting for energy storage arbitrage, while Maciejowska et al. [2] examined the relationship between statistical forecasting accuracy and economic performance.

The present study combines these considerations within a consistent historical evaluation framework. It compares forecast-driven battery scheduling strategies under identical physical constraints, examines the economic effects of illustrative operating costs, and evaluates risk-aware optimization using historically calibrated price scenarios.

Particular attention is given to the transferability of historical forecast-error distributions and the distinction between scenario-based risk measures and realized downside performance.

TABLE I
CHRONOLOGICAL DATA PARTITIONING

Period	Purpose
2012–2014	Initial model training
2015	Validation and selection
2012–2015	Final model training
2016–2017	Original out-of-sample test
2019	Later-period evaluation
2020	Later-period evaluation

The resulting analysis provides a comparative empirical assessment of the evaluated methods without claiming a new forecasting algorithm or optimization formulation.

III. DATA AND FORECASTING METHODOLOGY

A. Electricity Market Data

The study uses historical hourly day-ahead electricity prices from the German electricity market. Two data sources are employed to evaluate forecasting and trading performance across different historical periods.

The initial dataset covers January 9, 2012, to December 31, 2017, and is obtained from the publicly available EPEX-Germany benchmark dataset described in [6]. The electricity price series is extracted for the forecasting analysis.

Additional day-ahead electricity prices for 2019 and 2020 are obtained from the SMARD electricity market data platform [8], corresponding to the Germany/Luxembourg bidding zone.

The later datasets contain 8,760 hourly observations for 2019 and 8,784 for 2020. Their timestamps are converted to the Europe/Berlin time zone and chronologically ordered using unambiguous UTC timestamps.

The original benchmark and the later SMARD datasets are not fully homogeneous. Differences in bidding-zone configuration, timestamp representation, and historical market conditions are therefore considered when interpreting cross-period results.

B. Chronological Data Partitioning

A chronological partitioning strategy is adopted to prevent future target observations from entering the model training process.

Observations from 2012–2014 are used for initial training, while 2015 serves as the validation period for hyperparameter selection.

After model selection, the forecasting model is retrained using all available observations through December 2015. The resulting model is evaluated on 2016–2017, 2019, and 2020 without further retraining or hyperparameter adjustments.

Table I summarizes the experimental periods.

C. Feature Engineering

The forecasting model uses lagged electricity prices and calendar-based predictors.

For each hourly observation t, the feature vector is defined as

$$\mathbf{x}_t = [p_{t-24}, p_{t-48}, p_{t-168}, h_t, d_t, m_t, s_t, c_t], \tag{1}$$

where p_{t-k} denotes the electricity price located k records before observation t in the chronologically ordered dataset. Furthermore, h_t denotes the local hour of the day, d_t the day of the week, and m_t the calendar month.

The cyclic hour variables are defined as

$$s_t = \sin\left(\frac{2\pi h_t}{24}\right), \qquad c_t = \cos\left(\frac{2\pi h_t}{24}\right). \tag{2}$$

The lagged prices capture historical price dependencies, while the calendar variables represent recurring temporal patterns.

Features are constructed using positional shifts of the hourly price series. Therefore, the lag indices represent numbers of records rather than guaranteed differences in local clock time. This distinction is relevant during daylight-saving time transitions.

Observations without the required historical lags are excluded from forecasting evaluation.

D. Machine Learning Model

Electricity prices are predicted using a histogram-based gradient boosting regression model implemented with scikit-learn.

The forecasting function is represented by

$$\hat{p}_t = f_{\boldsymbol{\theta}}(\mathbf{x}_t), \tag{3}$$

where $\boldsymbol{\theta}$ denotes the learned model parameters.

Three candidate hyperparameter configurations are evaluated using the 2015 validation period. The configuration achieving the lowest validation Mean Absolute Error is selected.

The selected model uses a learning rate of 0.05, 150 boosting iterations, and a maximum of 15 leaf nodes per tree. Automatic early stopping is disabled, and a fixed random seed of 42 is used.

The final model is fitted using observations through December 2015 and remains unchanged throughout the subsequent evaluation periods.

E. Day-Ahead Information Availability

In day-ahead electricity trading, forecasts must be generated using information available before the relevant market gate closure. Accordingly, chronological separation between training and evaluation data must be complemented by an assessment of feature availability at the decision time.

The calendar predictors are known in advance. The price-based predictors use previously published day-ahead prices rather than contemporaneous realized prices.

Under the conventional day-ahead publication schedule, prices for a given delivery date are published on the preceding calendar day. Consequently, prices associated with delivery dates earlier than the target delivery date are generally available before the target day's auction.

A timestamp-based availability audit was performed for the 2019 and 2020 datasets. For the evaluated observations, all three lagged-price features refer to earlier delivery dates except for one observation in each year, associated with a daylight-saving time transition.

Neither exception belongs to a delivery day containing exactly 24 hourly records. Thus, the regular 24-hour delivery days retained in the later-period battery trading experiments satisfy this date-based availability criterion.

This audit does not constitute a complete reconstruction of historical market publication timestamps or auction gate-closure conditions.

F. Forecasting Benchmarks and Metrics

The machine learning forecasts are compared against a persistence-based naive benchmark defined as

$$\hat{p}_t^{\text{naive}} = p_{t-24}. \tag{4}$$

Forecasting accuracy is assessed using the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N}\sum_{t=1}^{N} |p_t - \hat{p}_t|\,, \tag{5}$$

and the Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N} (p_t - \hat{p}_t)^2}. \tag{6}$$

Both metrics are expressed in EUR/MWh.

For the later-period datasets, the first 168 hourly observations are used to initialize the lagged features. Consequently, the forecasting evaluation includes 8,592 observations in 2019 and 8,616 observations in 2020.

Forecasting performance is subsequently assessed from an economic perspective. Predicted electricity prices determine battery charging and discharging schedules, while realized market prices are used to calculate the corresponding trading profits.

G. Temporal and Data Limitations

The original historical benchmark is represented using a standardized sequence of 24 hourly observations per day, whereas the later SMARD datasets preserve actual daylight-saving time transitions.

As a result, observation-based price lags do not always correspond to the same local clock hours across the datasets.

The forecasting evaluation retains these observations, including daylight-saving transition periods. In contrast, the daily battery scheduling experiments exclude delivery dates containing fewer or more than 24 hourly observations.

The resulting differences in timestamp representation, together with changes in bidding-zone configuration and market conditions, limit the direct comparability of forecasting and trading performance across historical periods.

TABLE II
BATTERY MODEL PARAMETERS

Parameter	Value
Maximum charging power	1 MW
Maximum discharging power	1 MW
Energy capacity	2 MWh
Round-trip efficiency	90%
Scheduling horizon	24 h
Time resolution	1 h
Initial stored energy	0 MWh
Terminal stored energy	0 MWh

IV. BATTERY TRADING OPTIMIZATION

A. Battery Model and Operating Constraints

A battery energy storage system (BESS) is considered for price-taking arbitrage in the day-ahead electricity market.

The BESS has a rated charging and discharging power of 1 MW, an energy capacity of 2 MWh, and a round-trip efficiency of 90%.

Each optimization problem covers a single delivery day comprising 24 hourly intervals. The charging and discharging schedule is determined using forecasted electricity prices and remains fixed during the delivery day.

The decision variables are charging power P_t^{ch}, discharging power P_t^{dis}, stored energy E_t, and a binary operating-mode variable z_t.

The stored energy evolves according to

$$E_t = E_{t-1} + \eta_{\text{ch}} P_t^{\text{ch}} \Delta t - \frac{P_t^{\text{dis}} \Delta t}{\eta_{\text{dis}}}, \tag{7}$$

for $t = 1, \ldots, T$, where $T = 24$ and $\Delta t = 1$ h.

Symmetric charging and discharging efficiencies are assumed:

$$\eta_{\text{ch}} = \eta_{\text{dis}} = \sqrt{0.90}. \tag{8}$$

The operating constraints are

$$0 \le E_t \le E_{\max}, \tag{9}$$

$$0 \le P_t^{\text{ch}} \le P_{\max}(1 - z_t), \tag{10}$$

$$0 \le P_t^{\text{dis}} \le P_{\max} z_t, \tag{11}$$

$$z_t \in \{0, 1\}. \tag{12}$$

The binary variable prevents simultaneous charging and discharging, while allowing idle operation.

The initial and terminal stored energy levels are fixed at zero:

$$E_0 = E_T = 0. \tag{13}$$

This condition prevents energy carryover between consecutive delivery days and allows each day to be optimized independently. It is a simplifying assumption rather than a requirement of practical battery operation.

Table II summarizes the battery parameters.

B. Deterministic MILP Formulation

The deterministic optimization model uses the predicted day-ahead electricity prices $\hat{p}_t$ to determine the battery trading schedule.

The objective is to maximize forecasted gross trading profit:

$$\max \hat{\Pi}_{\text{gross}} = \sum_{t=1}^{T} \hat{p}_t \left(P_t^{\text{dis}} - P_t^{\text{ch}}\right) \Delta t. \tag{14}$$

The objective is optimized subject to the battery constraints (7)– (13).

Charging corresponds to electricity purchases, while discharging corresponds to electricity sales.

The resulting formulation is a mixed-integer linear programming (MILP) problem.

For historical backtesting, the schedule is optimized using forecasted prices, while realized gross profit is evaluated using observed market prices p_t:

$$\Pi_{\text{gross}} = \sum_{t=1}^{T} p_t \left(P_t^{\text{dis}} - P_t^{\text{ch}}\right) \Delta t. \tag{15}$$

A perfect-foresight benchmark is constructed by solving the same optimization problem using realized prices in the objective.

This oracle benchmark is not an implementable trading strategy. It represents an ex-post upper bound on gross arbitrage profit for the specified battery model and daily scheduling constraints.

C. Cost-Aware MILP Formulation

Gross-profit maximization does not explicitly account for battery degradation or transaction-related operating costs.

A cost-aware formulation is therefore introduced to examine how assumed variable costs influence battery dispatch decisions.

Let c_{deg} denote an illustrative degradation cost per MWh discharged, and let c_{trade} denote the transaction cost per MWh charged or discharged.

The forecasted net operating profit is

$$\begin{aligned} \hat{\Pi}_{\text{net}} = \sum_{t=1}^{T} \Big[& \hat{p}_t (P_t^{\text{dis}} - P_t^{\text{ch}}) \\ & - c_{\text{deg}} P_t^{\text{dis}} \\ & - c_{\text{trade}} (P_t^{\text{ch}} + P_t^{\text{dis}}) \Big] \Delta t. \end{aligned} \tag{16}$$

The cost-aware MILP maximizes $\hat{\Pi}_{\text{net}}$ subject to the same battery operating constraints as the deterministic model.

Realized net operating profit is calculated as

$$\begin{aligned} \Pi_{\text{net}} = \Pi_{\text{gross}} & - c_{\text{deg}} E_{\text{dis}} \\ & - c_{\text{trade}} (E_{\text{ch}} + E_{\text{dis}}), \end{aligned} \tag{17}$$

where total charged and discharged energy are defined as

$$E_{\text{ch}} = \sum_{t=1}^{T} P_t^{\text{ch}} \Delta t, \qquad E_{\text{dis}} = \sum_{t=1}^{T} P_t^{\text{dis}} \Delta t. \tag{18}$$

TABLE III
ILLUSTRATIVE OPERATING COST SCENARIOS

Scenario	c_{deg} (EUR/MWh)	c_{trade} (EUR/MWh)
No cost	0	0
Medium cost	5	0.50
High cost	10	1.00

Three illustrative operating-cost configurations are considered, as summarized in Table III.

These coefficients are illustrative scenario assumptions and are not calibrated to a specific commercial battery installation.

The formulation uses a linear throughput-based approximation of degradation-related costs. It does not explicitly model electrochemical aging, cycle-depth dependence, capital expenditure, or fixed operating costs.

Consequently, the resulting net profits should be interpreted as operating-profit estimates under the stated assumptions, rather than complete investment profitability measures.

D. Stochastic Optimization Framework

Deterministic optimization relies on a single forecasted electricity price trajectory. In practice, realized prices may differ from the forecast, potentially affecting trading profitability.

A scenario-based stochastic optimization framework is therefore introduced to represent forecast uncertainty.

For each delivery day, $S = 100$ electricity price scenarios are generated according to

$$p_{s,t} = \hat{p}_t + \varepsilon_{s,t}, \tag{19}$$

where $\varepsilon_{s,t}$ denotes a sampled historical forecasting error.

The residual trajectories are constructed from out-of-sample predictions for 2015, obtained using a forecasting model trained on observations through 2014.

For each optimization day, 100 complete daily residual trajectories are sampled without replacement from the historical residual pool.

Sampling entire daily trajectories preserves the empirical within-day dependence structure of the historical forecasting errors.

All scenarios share the same charging and discharging schedule. The formulation therefore represents a single-stage stochastic optimization problem with no scenario-dependent recourse during the delivery day.

The expected scenario profit is

$$\mathbb{E}[\Pi] = \frac{1}{S} \sum_{s=1}^{S} \Pi_s, \tag{20}$$

where scenario profit is

$$\Pi_s = \sum_{t=1}^{T} p_{s,t} (P_t^{\text{dis}} - P_t^{\text{ch}}) \Delta t. \tag{21}$$

E. Conditional Value-at-Risk

To incorporate downside-risk considerations, Conditional Value-at-Risk (CVaR) is introduced following [7].

Scenario loss is defined as the negative of scenario profit:

$$L_s = -\Pi_s. \tag{22}$$

For confidence level α, the empirical CVaR is represented by

$$\mathrm{CVaR}_\alpha(L) = \min_{\zeta,\mathbf{u}} \left[\zeta + \frac{1}{(1-\alpha)S} \sum_{s=1}^{S} u_s \right], \tag{23}$$

subject to

$$u_s \geq L_s - \zeta, \tag{24}$$

$$u_s \geq 0. \tag{25}$$

The auxiliary threshold ζ is unrestricted in sign.

The stochastic scheduling problem minimizes the negative expected scenario profit together with a weighted CVaR penalty:

$$\begin{aligned} \min \quad & -\mathbb{E}[\Pi] \\ & + \lambda \left[\zeta + \frac{1}{(1-\alpha)S} \sum_{s=1}^{S} u_s \right], \end{aligned} \tag{26}$$

subject to the battery operating constraints and CVaR auxiliary constraints.

The confidence level is set to $\alpha = 0.95$, and risk weights $\lambda \in \{0, 0.25, 1.0\}$ are evaluated.

For $\lambda = 0$, the optimization reduces to risk-neutral expected-profit maximization. Because all scenarios share a common schedule and profit is linear in electricity prices, this is equivalent to deterministic optimization using the mean scenario price trajectory.

For $\lambda > 0$, the objective balances expected profit against the empirical CVaR of scenario losses.

In particular, $\lambda = 1$ does not represent pure risk minimization, since expected profit remains part of the optimization objective.

The stochastic formulation optimizes gross trading profit and does not include the operating-cost coefficients introduced in the cost-aware model.

All strategies are evaluated ex post using realized market prices, independently of the scenario-based optimization objective.

The representativeness of the historical residual distribution and its transferability across evaluation periods are examined in the results and discussion.

V. Experimental Setup

A. Evaluation Protocol

The experimental framework evaluates the economic performance of forecast-driven battery trading strategies using historical day-ahead electricity market prices.

The forecasting model is trained on data through 2015 and evaluated on subsequent historical periods without retraining.

TABLE IV
Evaluation Samples for 2019 and 2020

Metric	2019	2020
Original hourly observations	8,760	8,784
Evaluated forecast hours	8,592	8,616
Eligible trading days	356	357

The original test period covers 2016–2017, while additional evaluations are conducted using the 2019 and 2020 Germany/Luxembourg market datasets.

The later periods provide an assessment of model performance under different market conditions and data characteristics.

For the 2019 and 2020 datasets, the first 168 hourly observations are used to initialize the lagged price features.

Battery trading experiments are conducted only on delivery days containing exactly 24 hourly observations with a complete set of local clock hours.

Days affected by daylight-saving time transitions are excluded from the daily optimization experiments.

Consequently, the 2019 and 2020 trading backtests contain 356 and 357 eligible days, respectively.

Table IV summarizes the evaluation samples.

Each battery optimization is solved independently for a single delivery day.

Initial and terminal stored energy are fixed at zero, and identical physical battery parameters are used across all trading strategies.

B. Trading Strategies and Benchmarks

Several trading strategies are evaluated to distinguish the effects of forecasting, operating costs, and uncertainty modeling.

The evaluated strategies are:

- *Fixed schedule:* A time-based benchmark constructed from historical electricity prices without using test-period forecasts.
- *Naive forecasting:* Battery dispatch obtained by optimizing the persistence-based electricity price forecast.
- *Deterministic ML:* Gross-profit maximization using machine learning price forecasts.
- *Risk-neutral stochastic:* Expected-profit maximization using the mean of sampled price scenarios.
- *CVaR-based stochastic:* Risk-aware optimization with risk weights of 0.25 and 1.0.
- *Cost-aware ML:* Net-profit maximization under illustrative degradation and transaction-cost assumptions.
- *Perfect foresight:* An ex-post benchmark that optimizes battery dispatch using realized electricity prices.

The fixed-schedule benchmark is constructed using historical prices from the original dataset through 2015.

For each hour of the day, the mean historical electricity price is calculated to obtain a representative 24-hour price profile.

The deterministic battery optimization model is then solved using this profile, producing a fixed charging and discharging schedule.

The resulting schedule is applied unchanged to each eligible delivery day in the 2019 and 2020 evaluations.

This benchmark therefore incorporates historically estimated intraday price patterns but does not respond to daily price forecasts.

All optimization-based strategies use identical physical battery constraints.

The perfect-foresight strategy is used exclusively as an ex-post upper benchmark for gross-profit evaluation and is not considered an implementable trading strategy.

C. Stochastic Scenario Construction

Forecast uncertainty is represented using historical daily residual trajectories from the 2015 calibration period.

The calibration model is trained using observations through 2014, and its forecasting errors are calculated using 2015 observations.

For each evaluated delivery day, 100 complete daily residual trajectories are randomly sampled without replacement from the 365 available historical days.

The sampled errors are added to the machine learning price forecast, producing 100 alternative daily price trajectories.

A fixed random seed of 42 is used for reproducibility.

The same sampled scenarios are used for all stochastic strategies evaluated on a given delivery day.

The CVaR confidence level is fixed at 95%, and risk weights of 0, 0.25, and 1.0 are considered.

Sampling complete residual trajectories preserves the empirical within-day dependence structure of the historical forecasting errors.

However, scenario generation is not conditioned on market regimes, seasons, or forecasted price levels.

Furthermore, the 2015 calibration period is also used for forecasting hyperparameter selection, while the residuals are generated by a model trained on a shorter historical period than the final forecasting model.

These characteristics limit the independence and potential transferability of the calibrated uncertainty distribution.

D. Economic Performance Metrics

The primary economic performance measure is realized trading profit.

For each delivery day d, the realized gross trading profit is denoted by Π_d^{gross}.

Net operating profit additionally subtracts the assumed degradation and transaction costs, as defined in Section IV.

For a strategy evaluated over D days, total realized profit is

$$\Pi_{\text{total}} = \sum_{d=1}^{D} \Pi_d. \tag{27}$$

Average daily profit is

$$\overline{\Pi} = \frac{1}{D} \sum_{d=1}^{D} \Pi_d. \tag{28}$$

The cumulative economic advantage of strategy A relative to benchmark B is defined as

$$\Delta\Pi_{A,B} = \sum_{d=1}^{D} \left(\Pi_{A,d} - \Pi_{B,d}\right). \tag{29}$$

Strategies are compared over identical eligible delivery dates.

Gross-profit comparisons and cost-aware net-profit comparisons are reported separately to avoid mixing different economic objectives.

The reported operating profits exclude capital expenditure, fixed operating expenses, financing costs, and other unmodeled market participation costs.

E. Realized Downside-Risk Metrics

In addition to total profitability, the analysis examines the distribution of realized daily trading profits.

The downside-risk metrics include:

- The number of days with negative realized daily profit.
- The minimum realized daily profit.
- The mean realized profit over the worst 5% of evaluated trading days.

For realized daily profits $\Pi_1, \ldots, \Pi_D$, let $\Pi_{(1)} \leq \cdots \leq \Pi_{(D)}$ denote the ordered observations.

The empirical worst-5% mean is defined as

$$\Pi_{\text{tail}} = \frac{1}{K} \sum_{i=1}^{K} \Pi_{(i)}, \tag{30}$$

where

$$K = \lceil 0.05D \rceil. \tag{31}$$

For the stochastic strategies, these metrics are calculated using realized gross trading profits under observed prices.

The realized worst-5% mean is distinct from the scenario-based CVaR quantity used during optimization.

The latter is calculated using the assumed price scenarios and is therefore an in-sample measure of modeled downside risk.

The two quantities describe different profit distributions and are not directly interchangeable.

F. Statistical Evaluation

Economic comparisons are conducted using paired daily profit differences over identical evaluation dates.

For strategies A and B, define

$$\delta_d = \Pi_{A,d} - \Pi_{B,d}. \tag{32}$$

The estimated mean daily advantage is

$$\overline{\delta} = \frac{1}{D} \sum_{d=1}^{D} \delta_d. \tag{33}$$

To account approximately for temporal dependence in daily profit differences, uncertainty is assessed using a moving-block bootstrap procedure.

TABLE V
OUT-OF-SAMPLE FORECAST ERRORS (EUR/MWH)

	MAE		RMSE	
Year	ML	Naive	ML	Naive
2019	7.205	9.128	12.316	14.602
2020	7.756	10.006	12.039	15.368

For the 2019 and 2020 evaluations, the procedure uses 14-day blocks and 10,000 bootstrap replications.

In each replication, contiguous blocks are sampled with replacement from the observed sequence of paired daily profit differences.

The concatenated blocks are truncated to the original sample length, and the mean profit difference is recalculated.

The 2.5th and 97.5th percentiles of the bootstrap distribution provide a 95% percentile confidence interval for the mean daily advantage.

Intervals containing zero are interpreted as providing insufficient evidence of a nonzero mean advantage under the adopted resampling assumptions.

The bootstrap analysis characterizes historical sampling uncertainty. It does not eliminate forecasting model risk, uncertainty arising from market regime changes, or limitations of the backtesting framework.

G. Computational Implementation

Forecasting and optimization experiments are implemented in Python.

The machine learning forecasting model uses the scikit-learn HistGradientBoostingRegressor.

Battery scheduling problems are formulated as mixed-integer linear programs and solved using the `scipy.optimize.milp` interface, which uses the HiGHS optimization solver.

A consistent set of physical battery parameters is applied across the compared strategies.

Random seeds are fixed for stochastic scenario generation and bootstrap resampling to support reproducibility.

Forecasts, realized trading profits, and statistical evaluation outputs are stored in CSV files for subsequent analysis and visualization.

The project repository includes the Python scripts and dependency requirements used in the experimental workflow.

VI. RESULTS AND DISCUSSION

A. Out-of-Sample Forecasting Accuracy

The forecasting model is evaluated on the 2019 and 2020 Germany/Luxembourg day-ahead electricity price datasets without retraining.

Table V compares its predictive performance against the persistence-based naive benchmark.

The machine learning model achieves lower forecasting errors in both evaluation years.

Relative to the naive benchmark, MAE decreases by approximately 21.07% in 2019 and 22.49% in 2020. RMSE is also lower in both periods, indicating improved performance under a metric that assigns greater weight to large prediction errors.

These results suggest that the model retains predictive value several years after its final training period, despite changes in market conditions and data characteristics.

Nevertheless, the remaining forecasting errors are economically relevant, and improved statistical accuracy does not necessarily imply higher trading profitability.

The following analysis therefore evaluates the realized economic performance of the resulting battery schedules.

TABLE VI
REALIZED GROSS TRADING PROFIT (EUR)

Strategy	2019	2020
Fixed schedule	15,001.46	17,038.66
Naive forecast	14,910.37	16,645.69
Deterministic ML	16,926.46	18,859.17
Perfect foresight	20,223.64	22,596.10

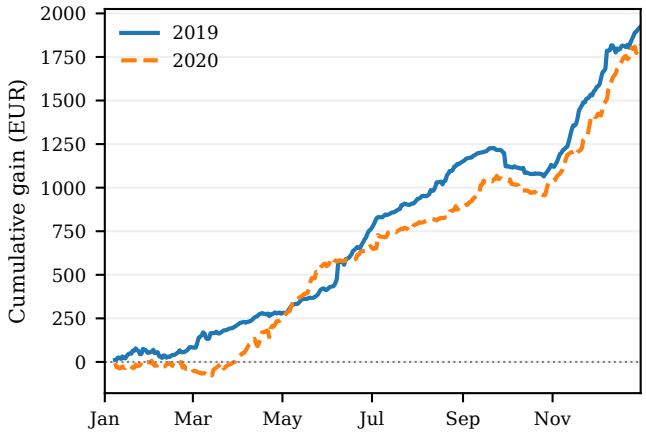


Fig. 1. Cumulative realized gross-profit advantage of ML-based battery scheduling over the fixed-schedule benchmark in 2019 and 2020.

B. Economic Value of ML-Based Trading

Table VI reports the realized gross trading profits of the principal battery scheduling strategies.

All results correspond to the 1 MW/2 MWh battery model and are expressed in EUR.

The deterministic ML strategy outperforms both implementable benchmarks in each evaluation year.

Relative to the fixed schedule, realized gross profit increases by approximately 12.83% in 2019 and 10.69% in 2020.

The corresponding ML profits represent approximately 83.70% and 83.46% of the perfect-foresight benchmark, respectively.

The oracle results represent ex-post upper bounds under the specified battery constraints and should not be interpreted as achievable day-ahead trading performance.

Fig. 1 illustrates the cumulative gross-profit advantage of ML-based scheduling relative to the fixed-schedule benchmark.

TABLE VII
REALIZED NET OPERATING PROFIT (EUR)

Year	Cost	Baseline	Cost-aware	Gain
2019	Medium	9,434.20	10,026.44	592.24
2019	High	1,941.94	4,616.44	2,674.50
2020	Medium	11,136.39	11,361.35	224.95
2020	High	3,413.62	6,161.12	2,747.49

The cumulative difference is positive at the end of both evaluation periods, although its evolution is not monotonic. Temporary declines indicate that the ML strategy does not outperform the benchmark on every trading day.

Paired moving-block bootstrap analysis provides additional evidence regarding the average daily economic advantage.

The 95% confidence intervals for the mean daily profit difference are [3.144, 7.583] EUR/day in 2019 and [3.153, 7.383] EUR/day in 2020.

Both intervals exclude zero under the adopted resampling assumptions.

Taken together, the results provide evidence that the evaluated ML-based forecasting approach improves realized gross arbitrage profit relative to the fixed-schedule benchmark over the considered historical periods.

However, these findings do not establish that comparable gains will persist under future market conditions.

C. Economic Impact of Operating Costs

The cost-aware MILP incorporates illustrative degradation and transaction costs directly into the battery scheduling objective.

Its performance is compared with the original gross-profit-maximizing ML schedule, with both schedules evaluated under identical operating-cost assumptions.

Table VII reports the resulting realized net operating profits.

Under the medium-cost assumptions, cost-aware scheduling improves realized net operating profit by approximately 592 EUR in 2019 and 225 EUR in 2020.

The 95% bootstrap confidence interval for the mean daily improvement is [0.824, 2.470] EUR/day in 2019.

For 2020, the corresponding interval is [-0.678, 1.679] EUR/day, which includes zero.

Thus, the evidence for a positive average medium-cost advantage is less conclusive in 2020.

Under the high-cost configuration, the realized net-profit improvements increase to approximately 2,675 EUR in 2019 and 2,747 EUR in 2020.

The corresponding 95% bootstrap confidence intervals for the mean daily advantage are [5.936, 9.266] EUR/day and [4.689, 10.260] EUR/day, respectively.

Both intervals exclude zero.

Fig. 2 shows the cumulative net-profit advantage of cost-aware scheduling under the high-cost assumptions.

The advantage generally increases over both evaluation periods, although temporary reductions are observed.

These findings indicate that explicitly accounting for variable operating costs can improve realized net profitability by discouraging transactions with insufficient expected margins.

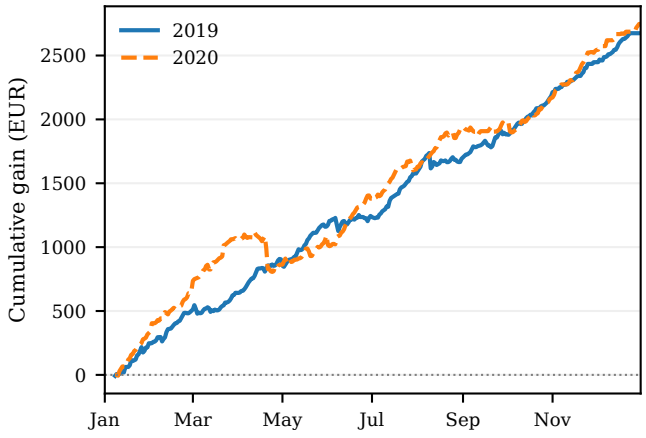


Fig. 2. Cumulative realized net-profit advantage of cost-aware scheduling over cost-blind scheduling under the illustrative high-cost configuration. Both schedules are evaluated using identical operating costs.

TABLE VIII
REALIZED GROSS PROFIT OF STOCHASTIC STRATEGIES (EUR)

Strategy	2019	2020
Deterministic ML	16,926.46	18,859.17
Risk-neutral	17,050.40	18,957.09
CVaR, $\lambda = 0.25$	16,831.81	18,736.81
CVaR, $\lambda = 1.00$	16,179.43	17,770.45

The magnitude of the improvement depends on the assumed cost coefficients and should not be interpreted as a complete battery lifecycle economic result.

D. Stochastic Optimization and CVaR

The stochastic strategies are evaluated using the same realized market prices and physical battery constraints as the deterministic ML benchmark.

Table VIII reports the realized gross profits under the different risk weights.

Risk-neutral scenario optimization produces small positive total profit differences of approximately 124 EUR in 2019 and 98 EUR in 2020.

However, the corresponding 95% bootstrap confidence intervals for the mean daily advantage include zero in both periods.

Because the risk-neutral formulation maximizes expected scenario profit, it is equivalent to deterministic optimization using the mean scenario price trajectory.

Its economic differences therefore reflect the adjustment introduced by the sampled scenario mean, rather than explicit downside-risk penalization.

For $\lambda = 0.25$, realized gross profit is slightly lower than under deterministic ML scheduling in both years. The bootstrap intervals for these differences also include zero.

Stronger risk penalization, with $\lambda = 1.00$, reduces realized gross profit by approximately 747 EUR in 2019 and 1,089 EUR in 2020.

The associated 95% bootstrap confidence intervals for the mean daily profit difference are [-2.585, -1.490] EUR/day and [-4.103, -2.006] EUR/day, respectively.

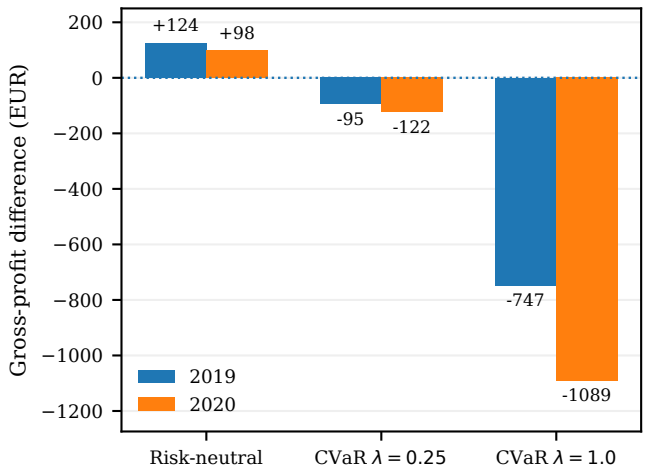


Fig. 3. Realized gross-profit differences between stochastic strategies and the deterministic ML benchmark in 2019 and 2020. Negative values indicate lower realized gross profit.

Both intervals are negative, indicating a realized gross-profit penalty under the adopted resampling assumptions.

Fig. 3 illustrates the differences in realized gross profit relative to deterministic ML scheduling.

Increasing the CVaR penalty reduces realized gross profit in both evaluation periods.

However, lower profitability does not necessarily imply that realized downside risk has been reduced.

Under $\lambda = 1.00$, the number of negative-profit days decreases from three to two in 2019 and from two to one in 2020, relative to deterministic ML scheduling.

Nevertheless, the realized worst-5% mean daily profit does not improve.

In 2019, this metric decreases from 4.33 EUR/day for deterministic ML scheduling to 3.98 EUR/day for $\lambda = 1.00$.

In 2020, the corresponding values are 8.54 and 8.19 EUR/day.

Therefore, the reduction in negative-profit days is not accompanied by an improvement in the evaluated lower-tail profit metric.

The results illustrate the difference between reducing modeled scenario losses and achieving better realized downside-risk outcomes.

E. Uncertainty Calibration and Distribution Shift

The stochastic framework uses historical forecasting residuals from 2015 to represent uncertainty during later evaluation periods.

To assess the transferability of this calibration distribution, the 2015 residuals are compared with realized forecasting errors from 2019 and 2020.

Table IX summarizes the principal diagnostics.

As shown in Fig. 4, forecast-error variability is substantially higher in both later evaluation periods than in the 2015 calibration sample.

TABLE IX
FORECAST RESIDUAL DIAGNOSTICS

Period	Bias (EUR/MWh)	Std. dev. (EUR/MWh)	Coverage (%)
2015	-1.154	8.297	–
2019	0.124	12.317	84.80
2020	-1.570	11.937	82.85

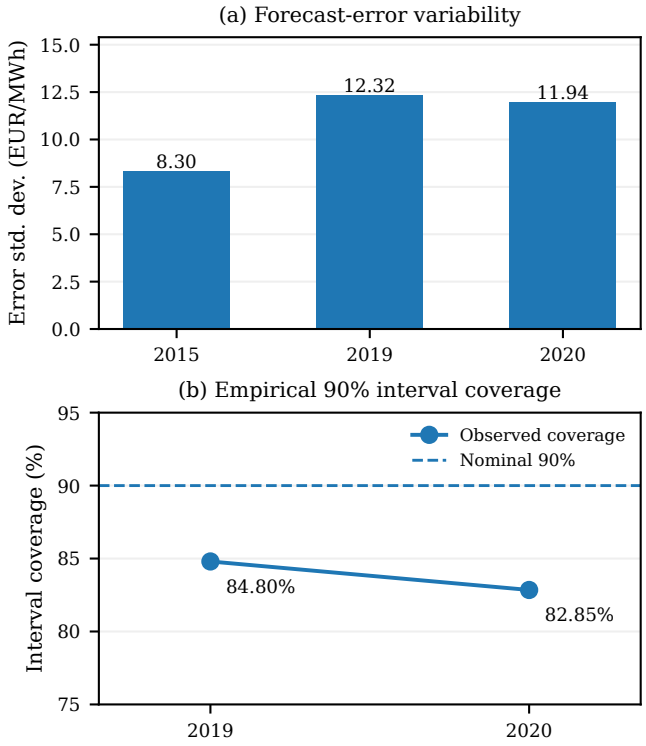


Fig. 4. Forecast uncertainty diagnostics: (a) residual standard deviation across historical periods and (b) empirical coverage of nominal 90% prediction-error intervals constructed from 2015 calibration residuals. The 2015 and later residuals correspond to models trained on different historical windows.

The residual standard deviation increases from 8.297 EUR/MWh in 2015 to 12.317 EUR/MWh in 2019 and 11.937 EUR/MWh in 2020.

These differences correspond to increases of approximately 48.4% and 43.9%, respectively.

Nominal 90% prediction-error intervals are constructed using the hourly 5th and 95th percentiles of the 2015 calibration residuals.

Their empirical coverage is 84.80% in 2019 and 82.85% in 2020, below the nominal level.

This undercoverage indicates that the fixed historical calibration distribution does not adequately describe the frequency of later forecast errors falling outside the constructed intervals.

An important qualification is that the 2015 residuals originate from a model trained through 2014, whereas the later residuals are generated by the final model trained through 2015.

Consequently, the observed differences cannot be attributed exclusively to changes in electricity market conditions.

Furthermore, the scenario generation procedure does not explicitly account for changing volatility regimes or conditional forecast-error distributions.

These findings suggest potential limitations in the uncertainty representation used by the stochastic optimization framework.

However, the hourly calibration diagnostics do not establish a causal explanation for the observed economic performance of the CVaR strategies.

F. Discussion and Limitations

The empirical results support three principal observations.

First, the machine learning forecasting model achieves lower prediction errors than the naive benchmark and is associated with higher realized battery arbitrage profits than the evaluated implementable trading benchmarks.

Second, incorporating illustrative variable operating costs directly into the scheduling objective can improve realized net profit, particularly when the assumed cost coefficients are relatively high.

Third, increasing the CVaR risk penalty does not necessarily improve realized downside-risk performance, even when it reduces the frequency of negative-profit trading days.

These findings should be interpreted within the limitations of the experimental framework.

The study assumes price-taking participation in the day-ahead market and does not explicitly model market impact, bid acceptance uncertainty, or detailed trading arrangements.

The battery model uses constant charging and discharging efficiencies and independent daily schedules with zero initial and terminal stored energy.

Consequently, multi-day energy shifting and more complex operational strategies are not represented.

The degradation-cost model is a linear approximation and does not capture detailed electrochemical aging or cycle-depth dependence.

The reported operating profits also exclude capital expenditure, fixed operating expenses, and financing costs.

Furthermore, the historical datasets differ in bidding-zone configuration and timestamp representation, limiting direct comparability across periods.

Finally, the stochastic optimization framework relies on a fixed historical residual distribution that may not adequately represent uncertainty under later market conditions.

The results should therefore be interpreted as historical evidence under explicitly defined modeling assumptions, rather than predictions of future commercial profitability.

VII. Conclusions and Future Work

This study examined the economic value of machine learning-based electricity price forecasting in combination with deterministic, cost-aware, and risk-aware battery arbitrage optimization.

The proposed framework was evaluated using historical German day-ahead electricity market data, with particular emphasis on the out-of-sample periods 2019 and 2020.

Three principal findings emerge from the empirical analysis.

First, the machine learning forecasting model achieved lower prediction errors than the naive benchmark and improved realized gross arbitrage profit relative to the fixed-schedule strategy.

For the considered 1 MW/2 MWh battery, deterministic ML-based scheduling increased gross trading profit by 12.83% in 2019 and 10.69% in 2020.

These results demonstrate the importance of complementing conventional forecasting metrics with decision-oriented economic evaluation.

Second, incorporating illustrative degradation and transaction costs directly into the scheduling objective improved realized net operating profit.

Under the high-cost configuration, cost-aware scheduling generated approximately 2,675 EUR and 2,747 EUR in additional net operating profit in 2019 and 2020, respectively, compared with cost-blind scheduling evaluated under identical costs.

This finding highlights the distinction between maximizing gross trading revenue and optimizing net operating profitability.

Third, stronger CVaR penalization reduced realized gross profit without improving the empirical worst-5% mean daily profit in either evaluation year.

Although the number of negative-profit days decreased slightly, the results did not establish a consistent improvement in realized downside-risk performance.

The uncertainty calibration analysis also revealed that the historical residual distribution provided insufficient coverage of later forecasting errors.

Empirical coverage of nominal 90% prediction-error intervals reached 84.80% in 2019 and 82.85% in 2020.

These findings highlight potential limitations of fixed historical error distributions in scenario-based risk optimization.

However, the calibration diagnostics do not establish a causal explanation for the observed CVaR performance.

A. Future Work

Several extensions could improve the realism and generalizability of the proposed framework.

One direction is adaptive uncertainty calibration using rolling historical windows, conditional residual distributions, or alternative scenario-generation methods.

Such approaches should be evaluated through strictly chronological validation and independent out-of-sample testing.

Another direction is the extension of independent daily scheduling to rolling-horizon or multi-day optimization, allowing energy carryover between delivery days.

More detailed battery degradation models could also be incorporated to represent cycle-depth dependence, state-of-charge effects, and long-term battery aging.

Finally, future research could consider additional electricity market mechanisms, practical trading constraints, and uncertainty in the execution of market decisions.

Overall, the findings emphasize that forecasting accuracy alone is insufficient to characterize the economic value of battery trading strategies.

Realized performance also depends on the optimization objective, operating-cost assumptions, and the quality of the uncertainty representation used in risk-aware decision-making.

REFERENCES

[1] L. Sang, Y. Xu, H. Long, Q. Hu, and H. Sun, "Electricity price prediction for energy storage system arbitrage: A decision-focused approach," *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 2822–2832, 2022.

[2] K. Maciejowska, A. Lipiecki, and B. Uniejewski, "Statistical and economic evaluation of forecasts in electricity markets: Beyond rmse and mae," *Energy Conversion and Management*, vol. 356, p. 121408, 2026.

[3] B. Xu, J. Zhao, T. Zheng, E. Litvinov, and D. S. Kirschen, "Factoring the cycle aging cost of batteries participating in electricity markets," in *2018 IEEE Power and Energy Society General Meeting (PESGM)*, 2018.

[4] P. L. Camuñas García-Miguel, J. Alonso-Martinez, S. Arnaltes Gómez, and J. L. Rodríguez-Amenedo, "Impact of risk measures and degradation cost on the optimal arbitrage schedule for battery energy storage systems," *International Journal of Electrical Power and Energy Systems*, vol. 157, p. 109883, 2024.

[5] R. Weron, "Electricity price forecasting: A review of the state-of-the-art with a look into the future," *International Journal of Forecasting*, vol. 30, no. 4, pp. 1030–1081, 2014.

[6] J. Lago, G. Marcjasz, B. De Schutter, and R. Weron, "Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark," *Applied Energy*, vol. 293, p. 116983, 2021.

[7] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *The Journal of Risk*, vol. 2, no. 3, 2000.

[8] Bundesnetzagentur, "SMARD market data: Wholesale electricity prices," https://www.smard.de/home/marktdaten, germany/Luxembourg day-ahead prices. Accessed: October 27, 2026.